

Fuel Rail

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

 ST3096-A	Brush, Fuel Injector 310-205
 ST3095-A	Remover, Fuel Injector 310-206
 ST3097-A	Seal Installer, Fuel Injector 310-207
 ST3266-A	Sizer, Teflon® Seal 303-1567
 ST3089-A	Slide Hammer 307-005 (T59L-100-B) or equivalent

Material

Item	Specification
Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A
Motorcraft® Orange Antifreeze/Coolant Concentrated VC-3-B (US); CVC-3-B2 (Canada)	WSS-M97B44-D

Removal

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: Do not carry personal electronic devices such as cell phones, pagers or audio equipment of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

WARNING: When handling fuel, always observe fuel handling precautions and be prepared in the event of fuel spillage. Spilled fuel may be ignited by hot vehicle components or other ignition sources. Failure to follow these instructions may result in serious personal injury.

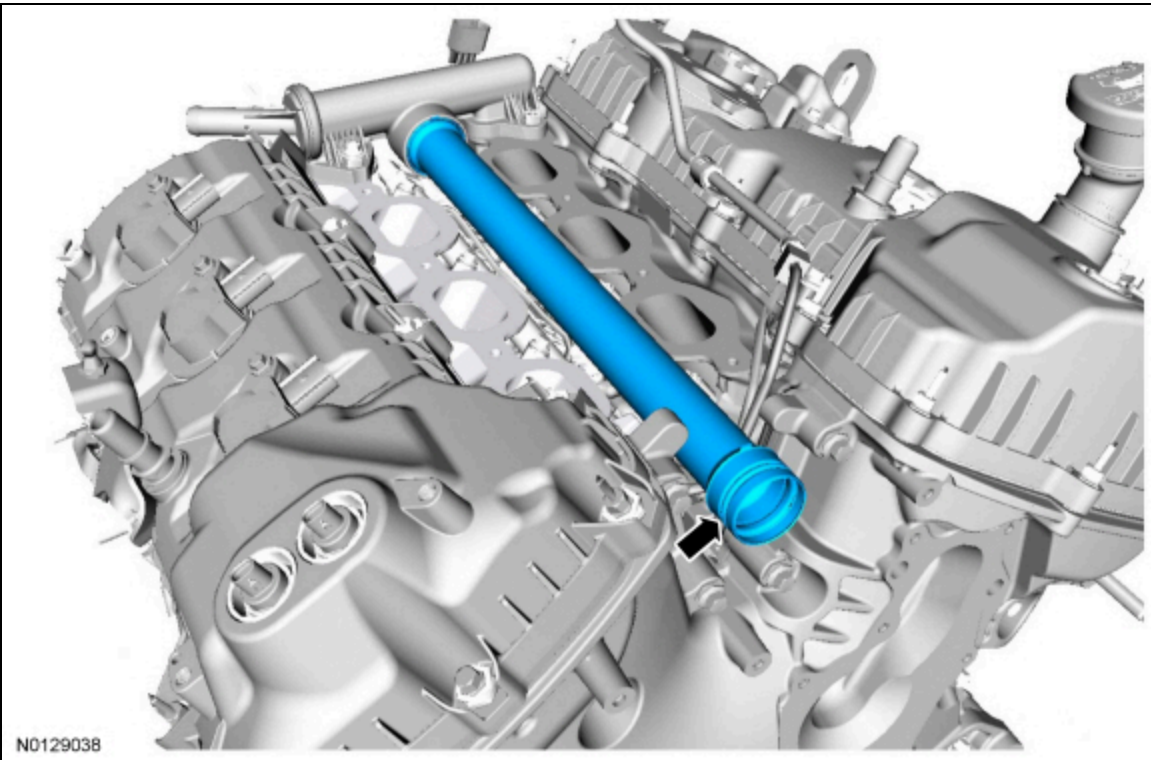
WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

WARNING: Clean all fuel residue from the engine compartment. If not removed, fuel residue may ignite when the engine is returned to operation. Failure to follow this instruction may result in serious personal injury.

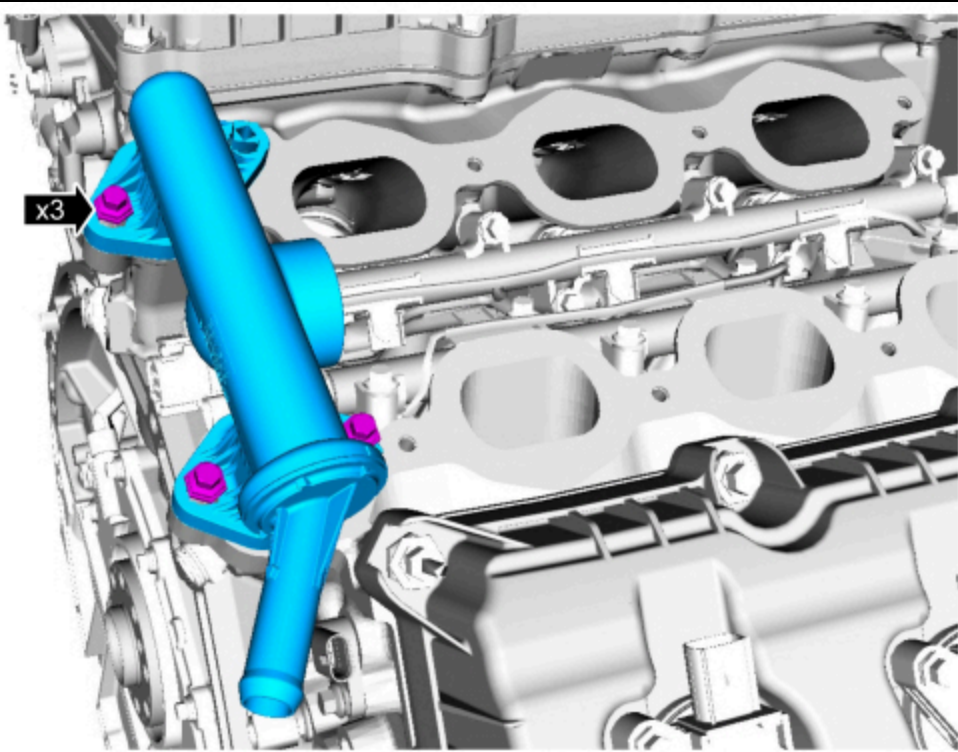
WARNING: Always disconnect the battery ground cable at the battery when working on an evaporative emission (EVAP) system or fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these instructions may result in serious personal injury.

NOTE: A clean working environment is essential to prevent dirt or foreign material contamination.

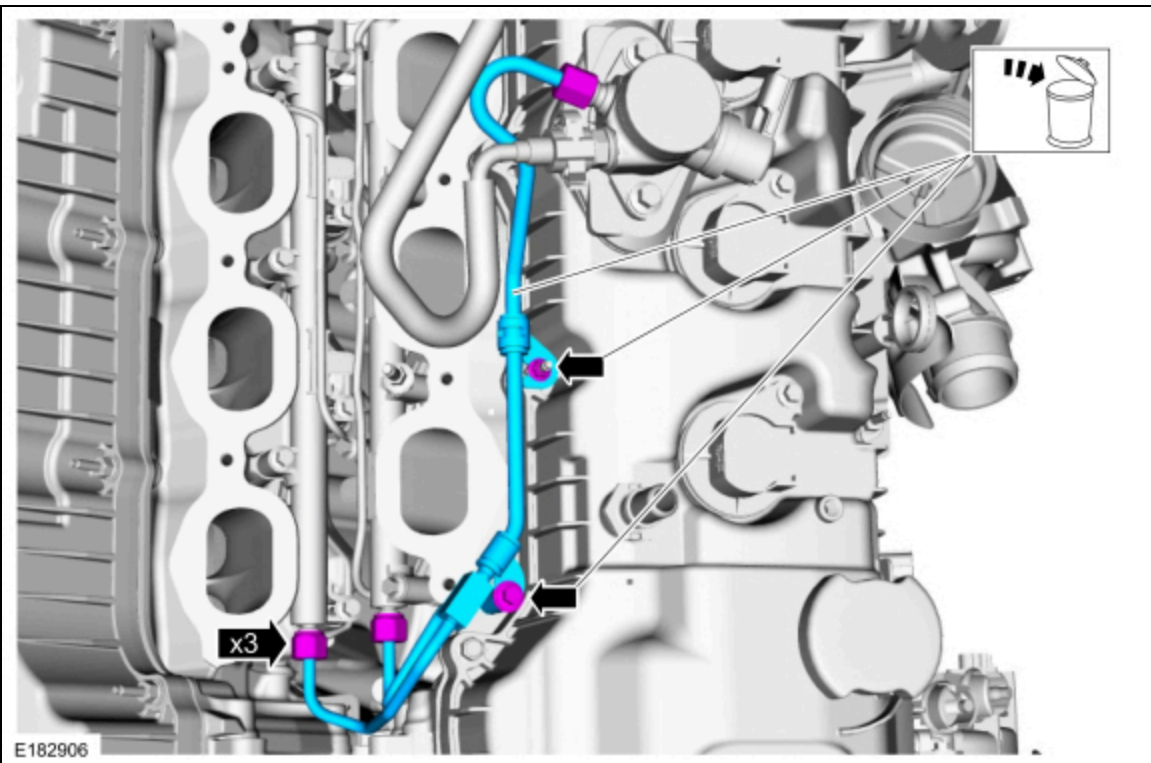
1. Release the fuel system pressure. For additional information, refer to Section 310-00.
2. Remove the intake manifold. For additional information, refer to Section 303-01A.
3. Remove the thermostat housing. For additional information, refer to Section 303-03.
4. Remove the coolant inlet pipe.
 - Remove and discard the coolant inlet pipe O-ring seals.



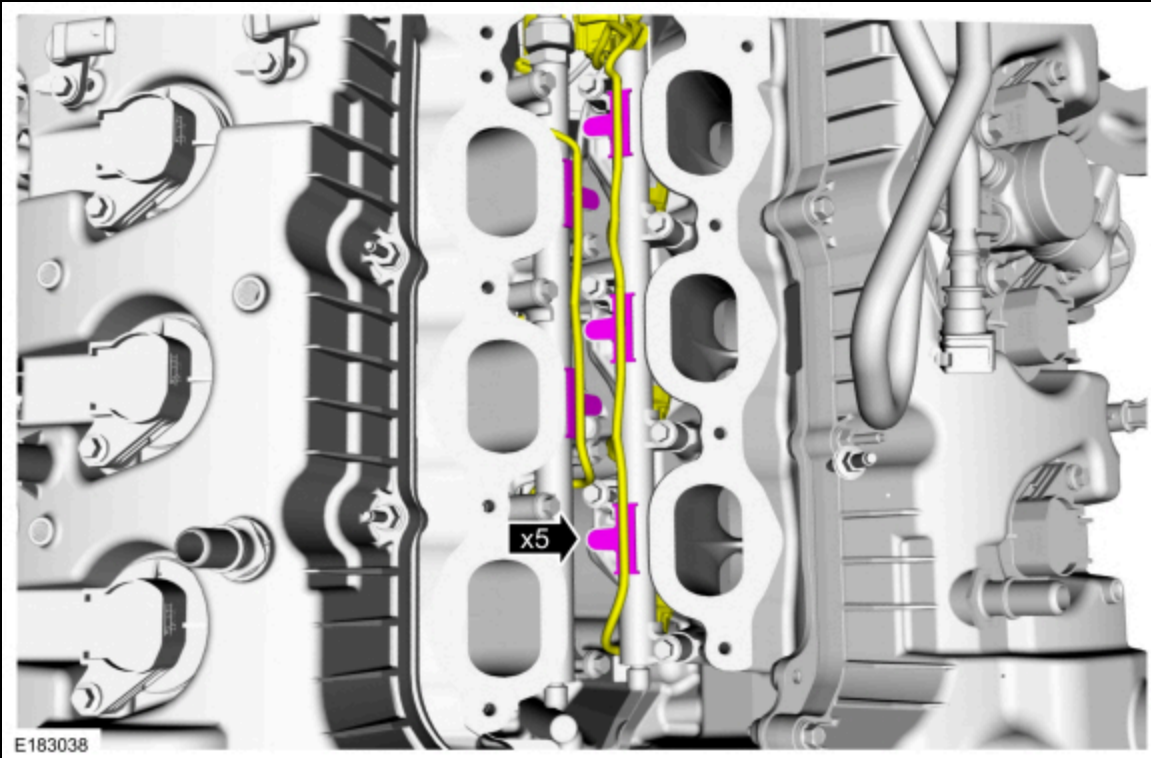
5. Remove the 3 bolts and position aside the coolant crossover assembly.
 - Remove and discard the 2 gaskets.



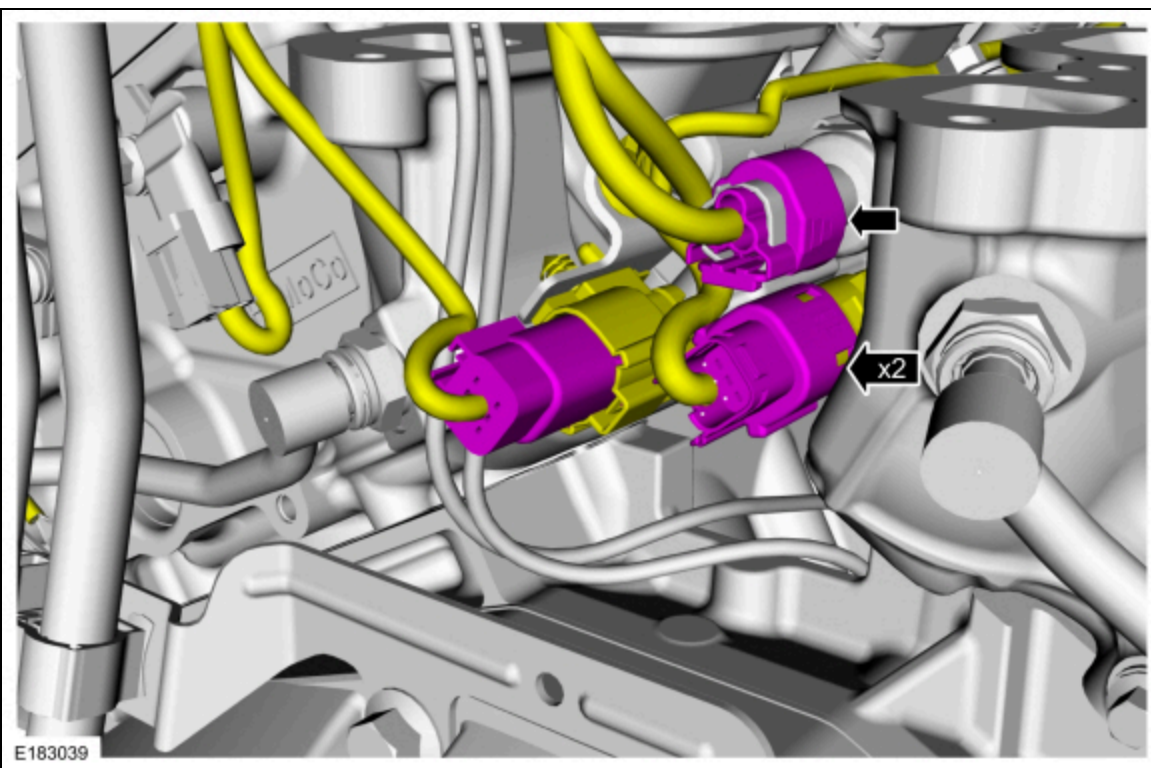
6. Remove and discard the high pressure fuel tube bracket nut and the high pressure fuel tube bracket bolt. Disconnect the high pressure fuel tube flare nuts, remove and discard the high pressure fuel tube.



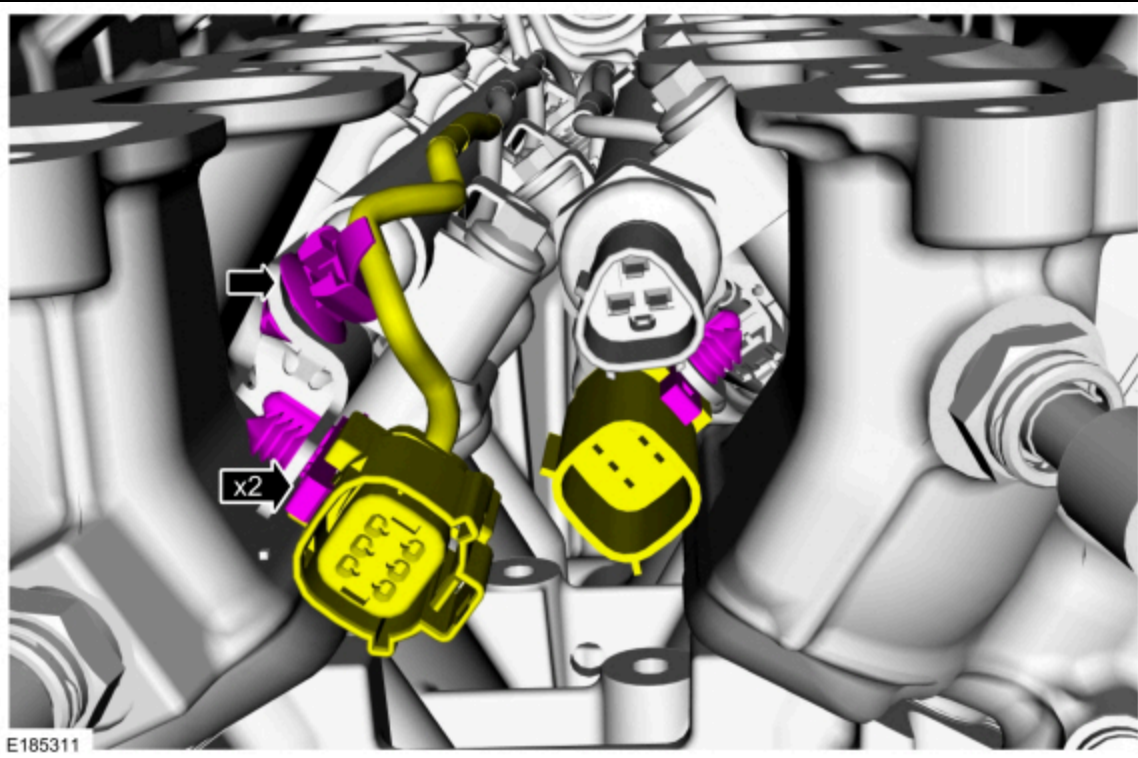
7. Detach the fuel injector wire harness retainers from the fuel rail.



8. Disconnect the fuel injector wire harness connectors and the fuel rail pressure sensor wire connector.



9. Detach the fuel injector wire harness electrical connector retainers and the fuel injector wire harness retainer from the fuel rail.



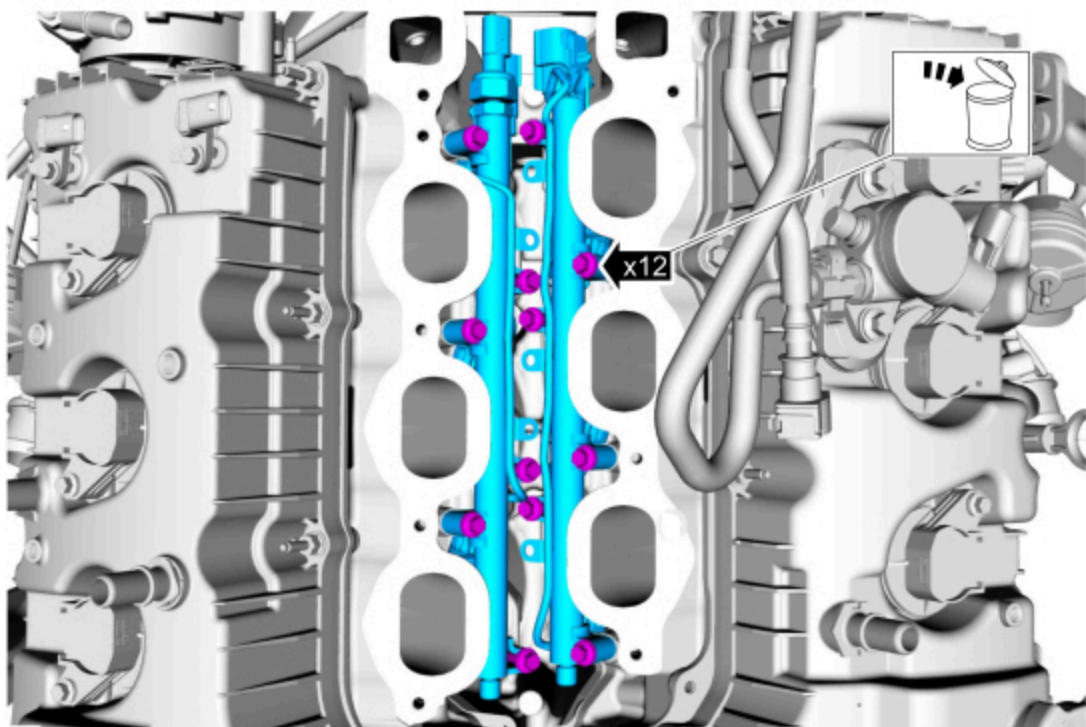
10. NOTICE: Pull out the fuel rails in the direction of the fuel injector axis or damage may occur to the fuel injectors.

NOTE: Use compressed air and remove any dirt or foreign material from the cylinder head, block and general surrounding area of the fuel rail and injectors.

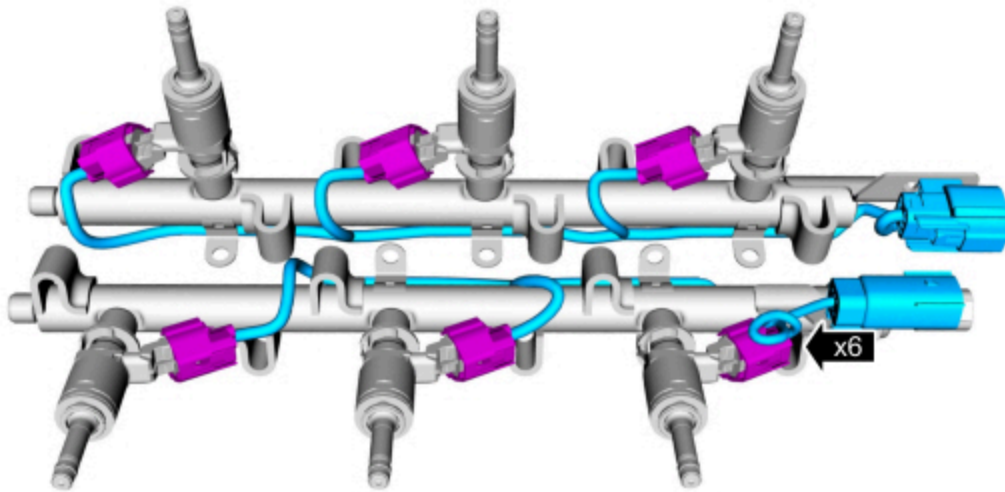
NOTE: When removing the fuel rails, the fuel injectors may remain in the cylinder heads and require the use of a Fuel Injector Remover tool to extract. Wiggling the injector by hand to break it loose may allow the injector to be removed by hand.

NOTE: Should any of the fuel injectors remain in the cylinder heads, then the fuel injector electrical connectors will need to be disconnected as the fuel rail is being removed.

Remove and discard the fuel rail bolts. Remove the fuel rails.

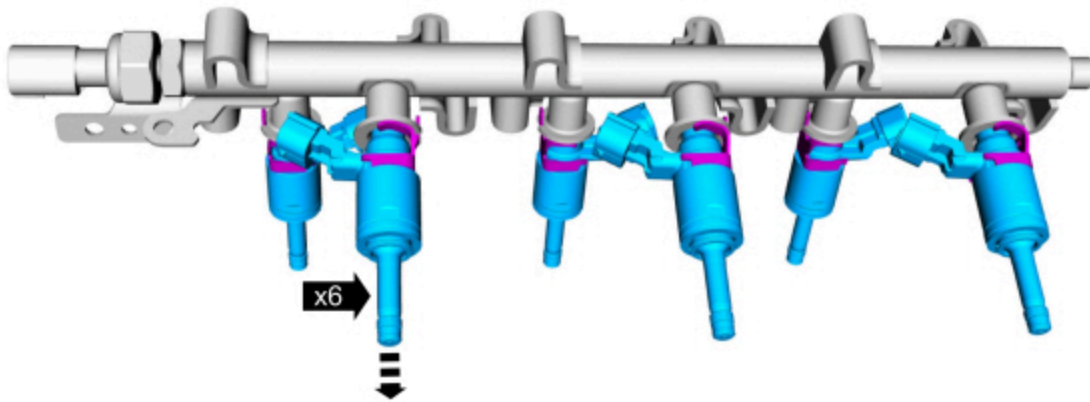


11. Disconnect the fuel injector electrical connectors.



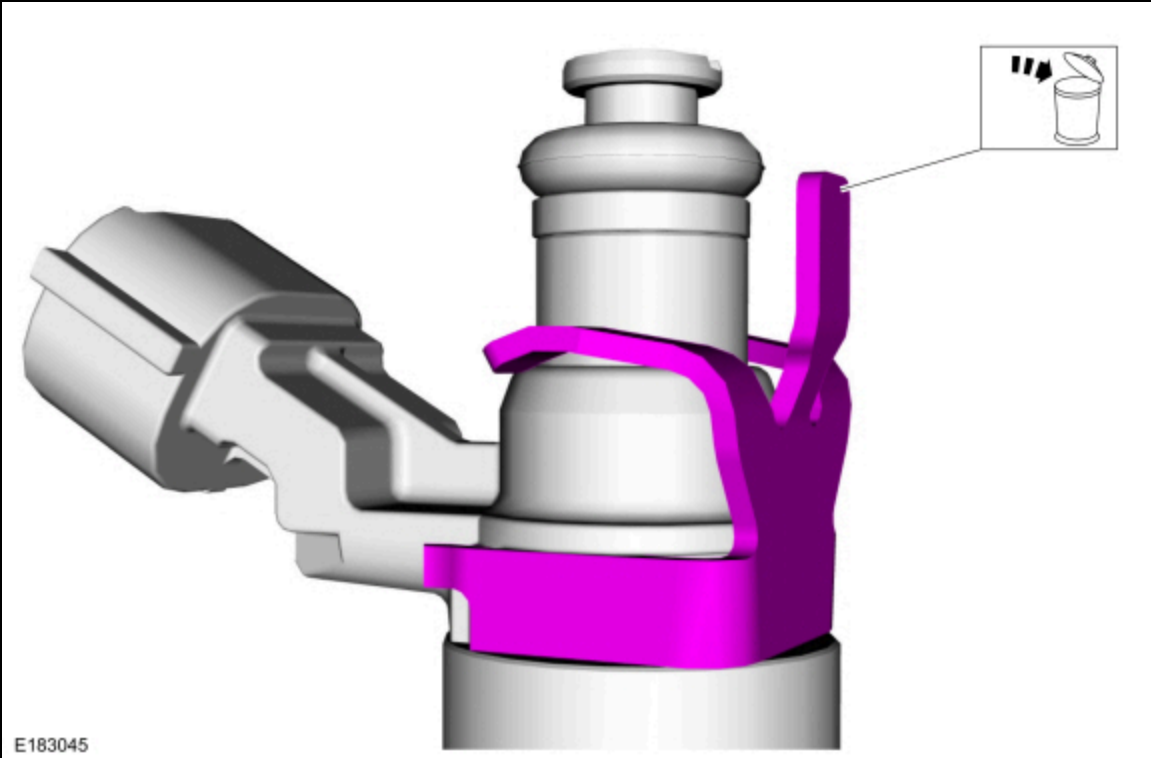
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12. Remove the fuel injectors from the fuel rail.



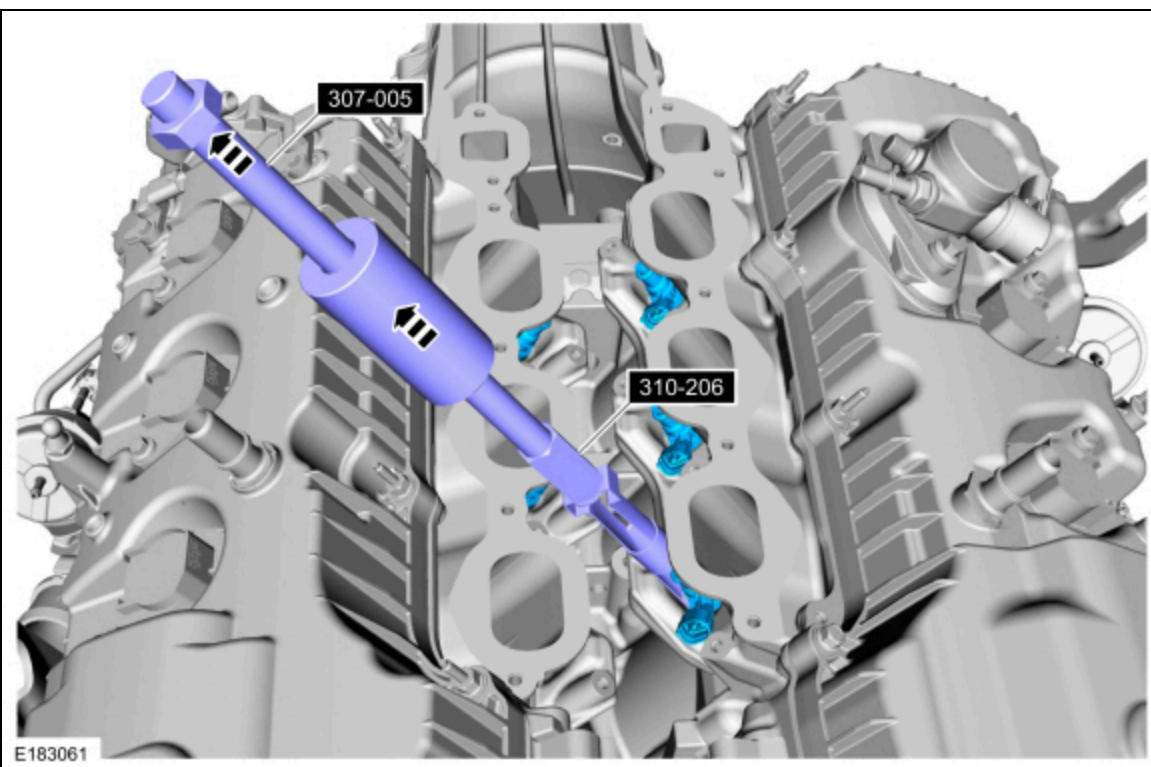
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13. Remove and discard the fuel injector clips.



14. **NOTICE:** Use minimal force to remove the fuel injectors that remained in the cylinder head with the Fuel Injector Remover tool or damage to the fuel injector assembly may occur.

Using the Fuel Injector Remover tool, remove any of the fuel injectors that remained in the cylinder head. Wiggling the injector by hand to break it loose may allow the injector to be removed by hand. Special Tool(s): Slide Hammer 307-005 (T59L-100-B) or equivalent and Remover, Fuel Injector 310-206..

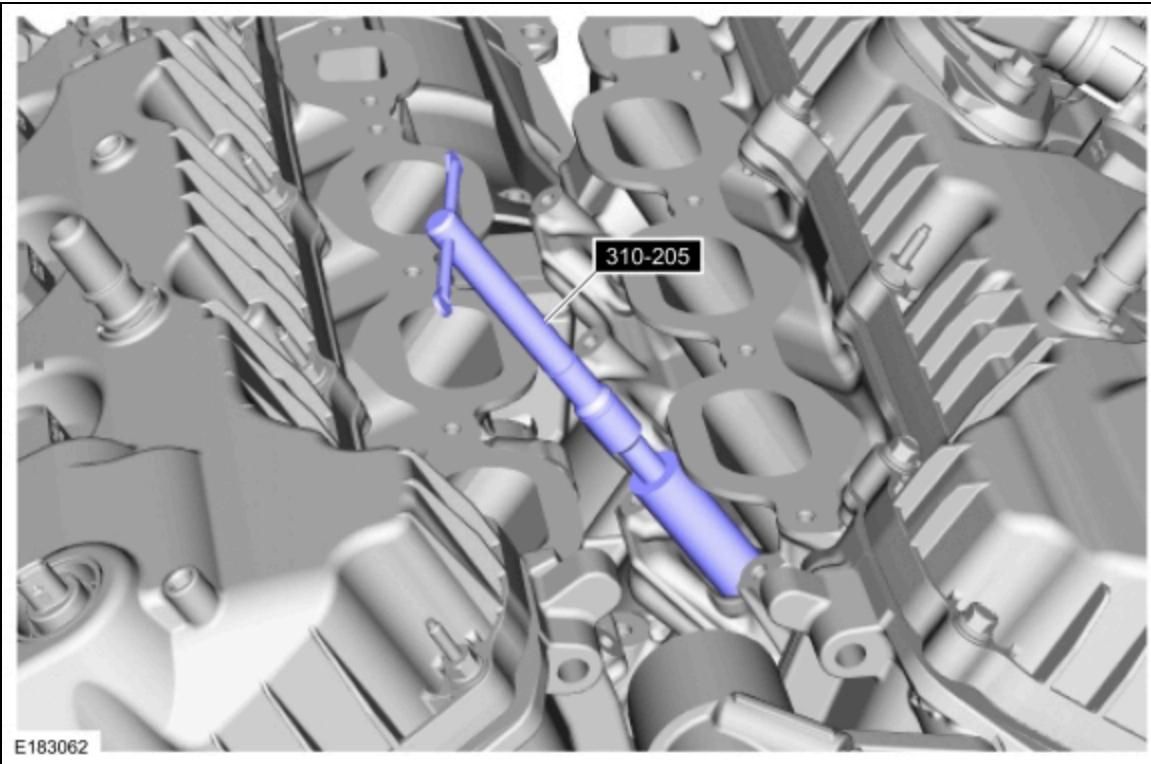


15. **NOTE:** Make sure to thoroughly clean any residual fuel or foreign material from the cylinder head, block and the general surrounding area of the fuel rails and injectors.

NOTE: Do not use compressed air to clean the tip of the fuel injector.

NOTE: Do not use a brush to clean the tip of the fuel injector.

Using the Fuel Injector Brush, clean the fuel injector bore in the cylinder head. Special Tool(s): Brush, Fuel Injector 310-205.



Installation

NOTE: A clean working environment is essential to prevent dirt or foreign material contamination.

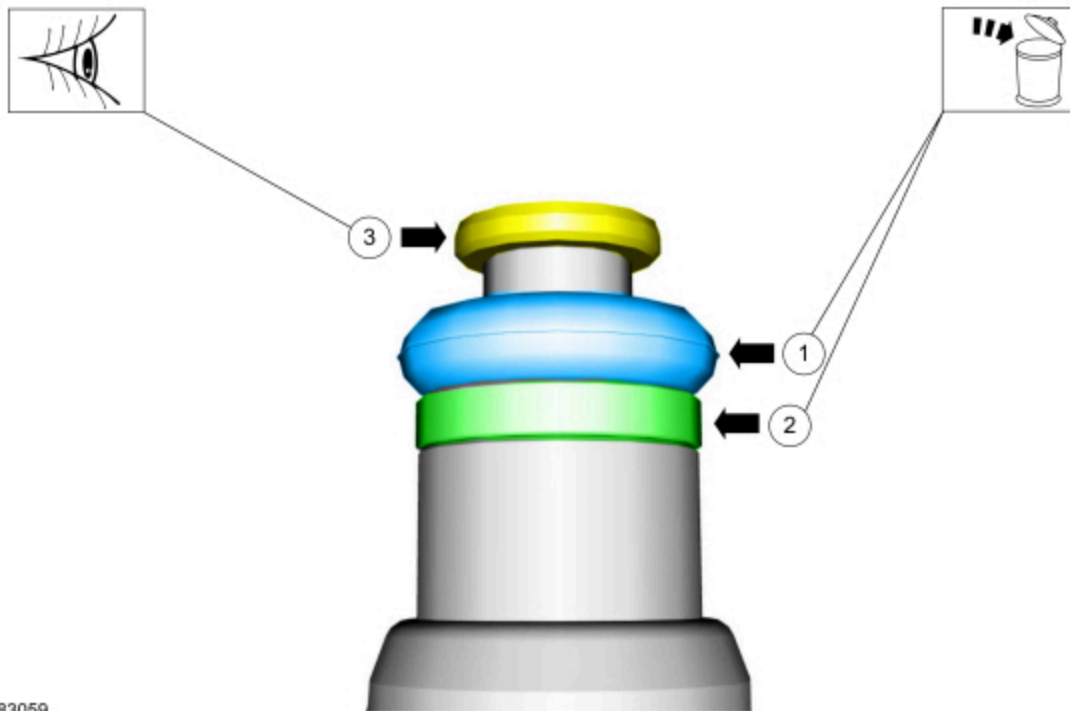
1.

1. Remove and discard the fuel injector O-rings.

2. **NOTICE:** Note the correct orientation of the fuel injector support rings for correct installation of the fuel injector support rings.

Remove and discard the fuel injector support rings.

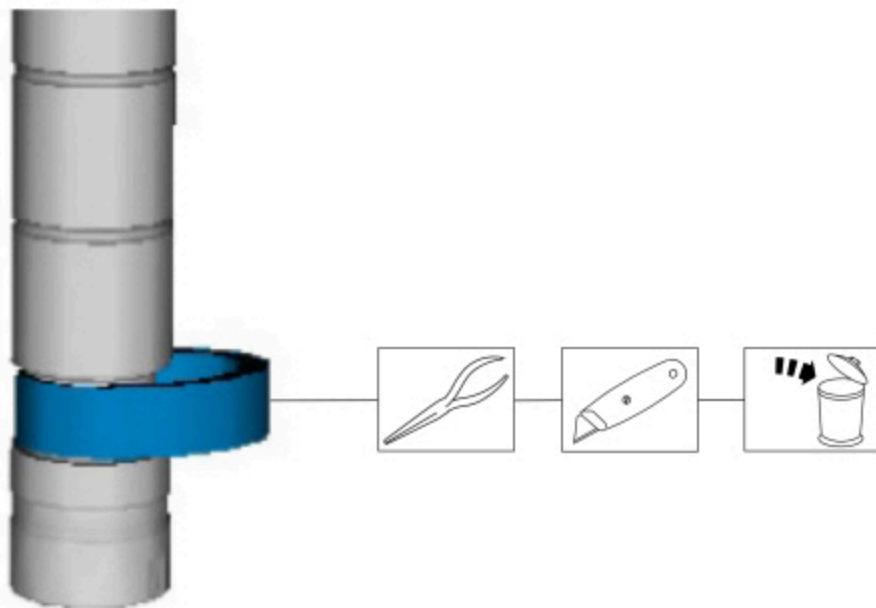
3. Inspect the inlet stopper, if it is damaged or comes off discard the inlet stopper. The inlet stopper is not necessary for proper function. Ensure that any debris from the inlet stopper are removed. If the inlet stopper is intact and undamaged do not remove the inlet stopper from the fuel injector.



2. **NOTICE:** Do not attempt to cut the lower Teflon® seal without first pulling it away from the fuel injector or damage to the injector may occur.

NOTICE: Use care when removing the lower Teflon® seals, not to scratch, nick or gouge the fuel injectors.

1. Pull the lower Teflon® seal away from the injector.
2. Carefully cut, remove and discard the lower fuel injector Teflon® seals.



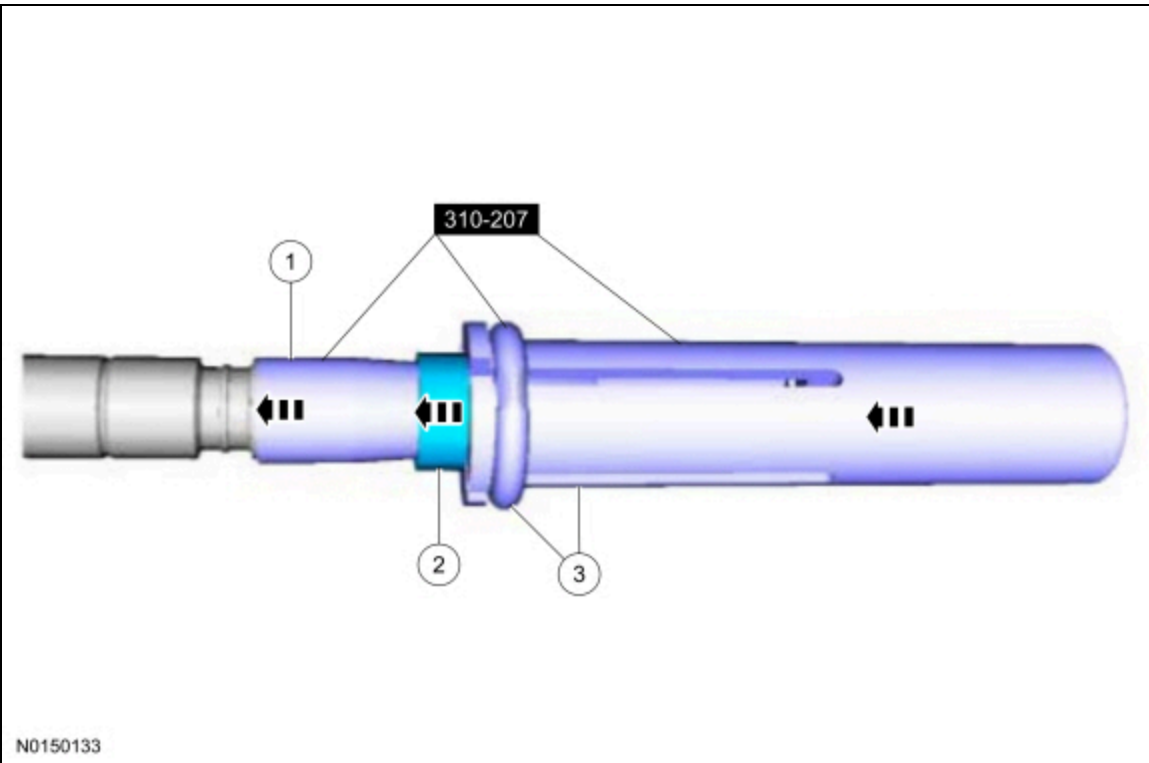
3. **NOTE:** Do not lubricate the new lower Teflon® fuel injector seals.

1. Install the Arbor on the fuel injector tips. Special Tool(s): Seal Installer, Fuel Injector 310-207.
2. **NOTICE:** Once the Teflon® seal is installed on the Teflon® Seal Guide 310-207 (part of the Fuel Injector Seal Installer), it should immediately be installed onto the fuel injector to avoid excessive expansion of the Teflon® seal.

NOTE: Make sure that new lower fuel injector Teflon® seals are installed.

Install the new Teflon® seals onto the Arbor, using the Pusher Tool (part of the Fuel Injector Seal Installer), slide the Teflon® seals along the Arbor. Special Tool(s): 310-207.

3. Using the Pusher Tool 310-207 (part of the Fuel Injector Seal Installer), slide the Teflon® seals off of the Teflon® Seal Guide and into the groove on the fuel injectors. Special Tool(s): Seal Installer, Fuel Injector 310-207.

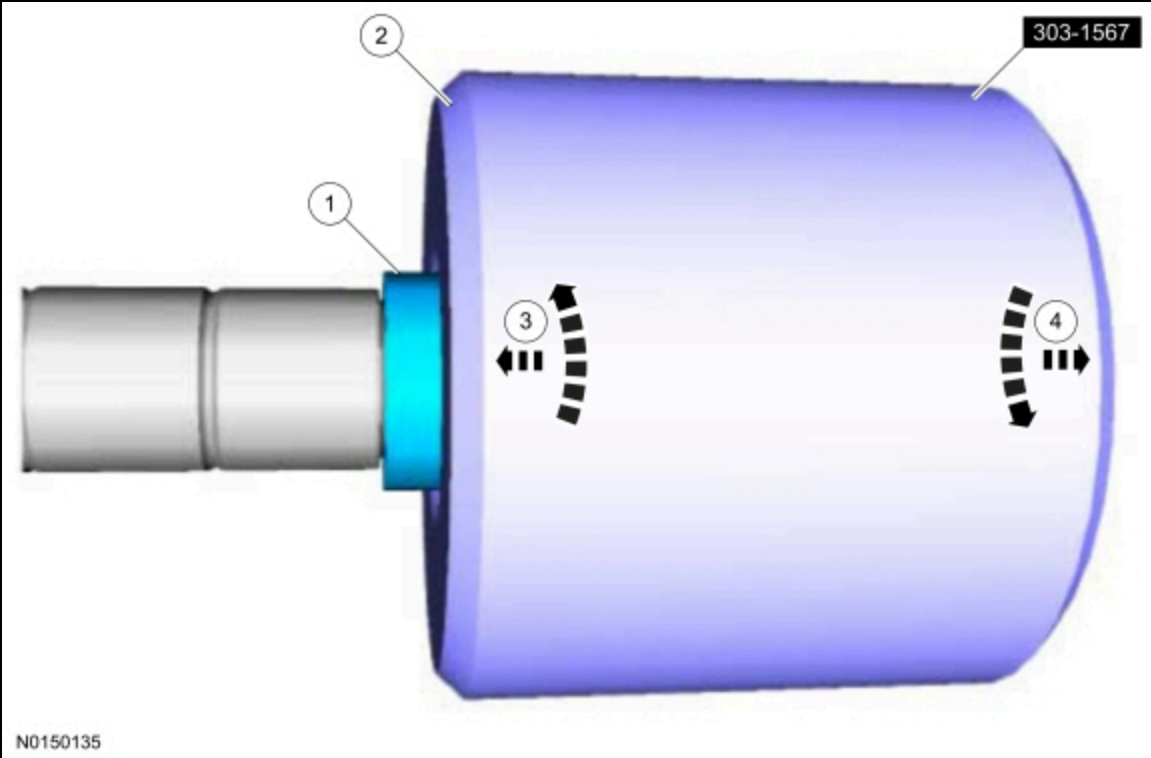


4. **NOTICE:** Install the fuel injectors into the cylinder head within 15 minutes of sizing the seals due to Teflon® seal expansion.

1. **NOTE:** Make sure the Teflon® seal is fully seated in the groove on the fuel injector before sizing the Teflon® seal.

Some Teflon® seal massaging with your fingers before the Teflon® seal sizer tool is installed will aid in installing the Teflon® seal sizer tool.

2. Position the Teflon® seal sizer tool 303-1567 with the larger opening towards the Teflon® seal. Push while turning the Teflon® seal sizer tool 180 degrees. Special Tool(s): Sizer, Teflon® Seal 303-1567.
3. Once the Teflon® seal sizer tool 303-1567 is installed, check and make sure the Teflon® seal is in the sizing portion of the Teflon® seal sizer tool.
4. After one minute, turn the Teflon® seal sizer tool back 180 degrees and remove.



5. **NOTE:** Make sure that new upper fuel injector O-ring seals, upper fuel injector O-ring seal support rings and fuel injector retainer clips are installed.

1. **NOTICE:** The new the fuel injector support rings must be installed in the correct orientation with the larger diameter facing up against the fuel injector O-ring seals. Improperly installed support rings may cause the fuel system to leak.

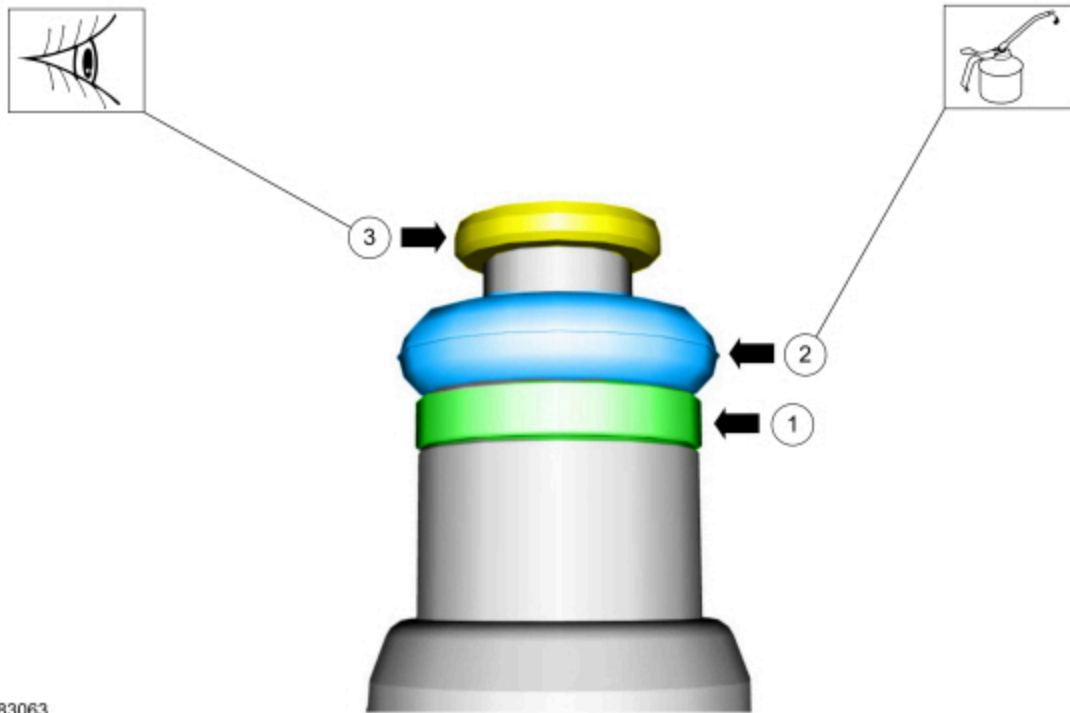
Install the new the fuel injector support rings.

2. **NOTICE:** Use fuel injector O-ring seals that are made of special fuel-resistant material. The use of ordinary O-ring seals may cause the fuel system to leak. Do not reuse the O-ring seals.

NOTICE: Do not lubricate the new lower Teflon® fuel injector seals.

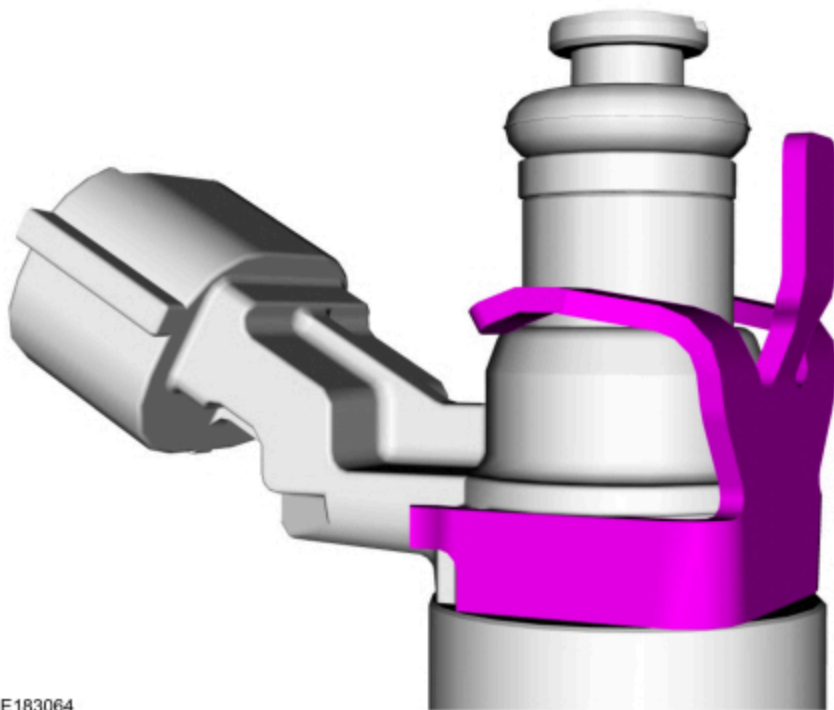
Install the new fuel injector O-rings and lubricate with clean engine oil. Material: Motorcraft® SAE 5W-30 Premium Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada); or equivalent.

3. Inspect the inlet stopper, if it is damaged or comes off discard the inlet stopper. The inlet stopper is not necessary for proper function. Ensure that any debris from the inlet stopper are removed. If the inlet stopper is intact and undamaged do not remove the inlet stopper from the fuel injector.



6. **NOTE:** Make sure that new fuel injector clips are installed.

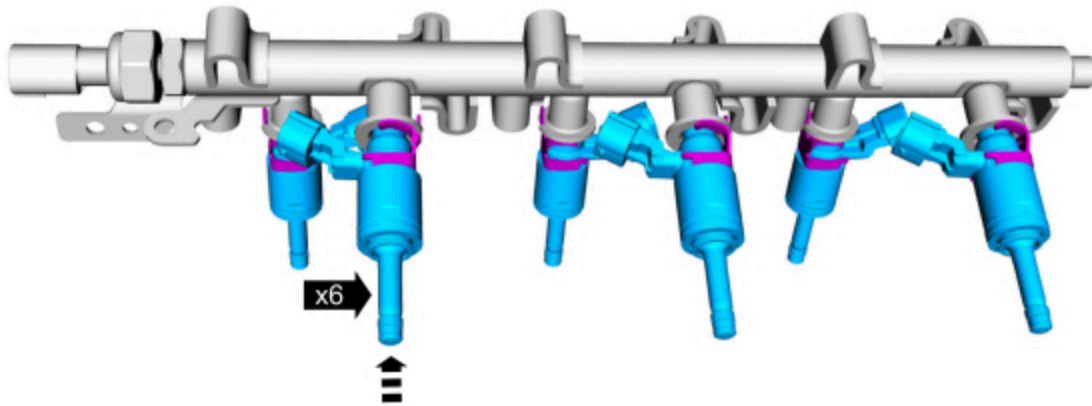
Install the new the fuel injector clips.



7. **NOTICE:** The ~~FRP~~ **FRP** sensor must be replaced if it is removed from the fuel rail. Refer to Section 303-14.

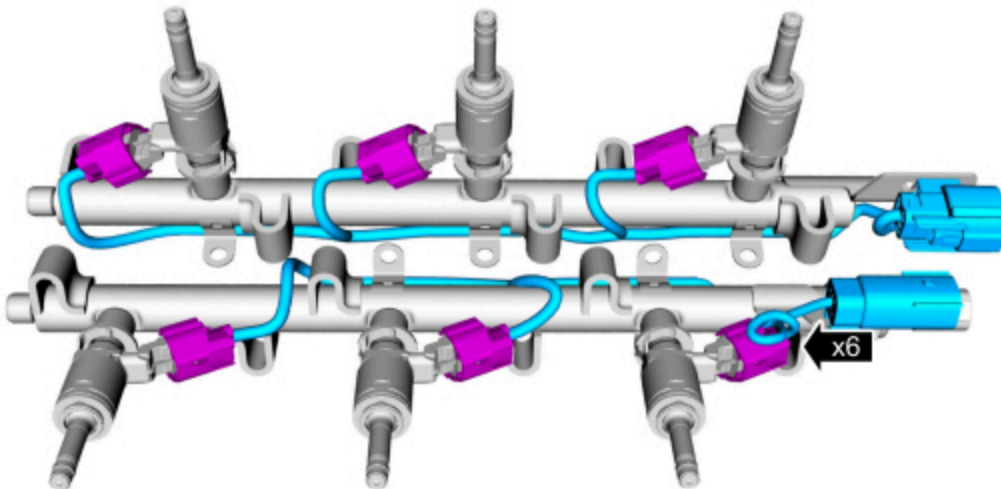
NOTE: The anti-rotation finger of the fuel injector clip must slip into the groove of the fuel rail cup.

Install the fuel injectors to the fuel rail.



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8. Connect the fuel injector electrical connectors.



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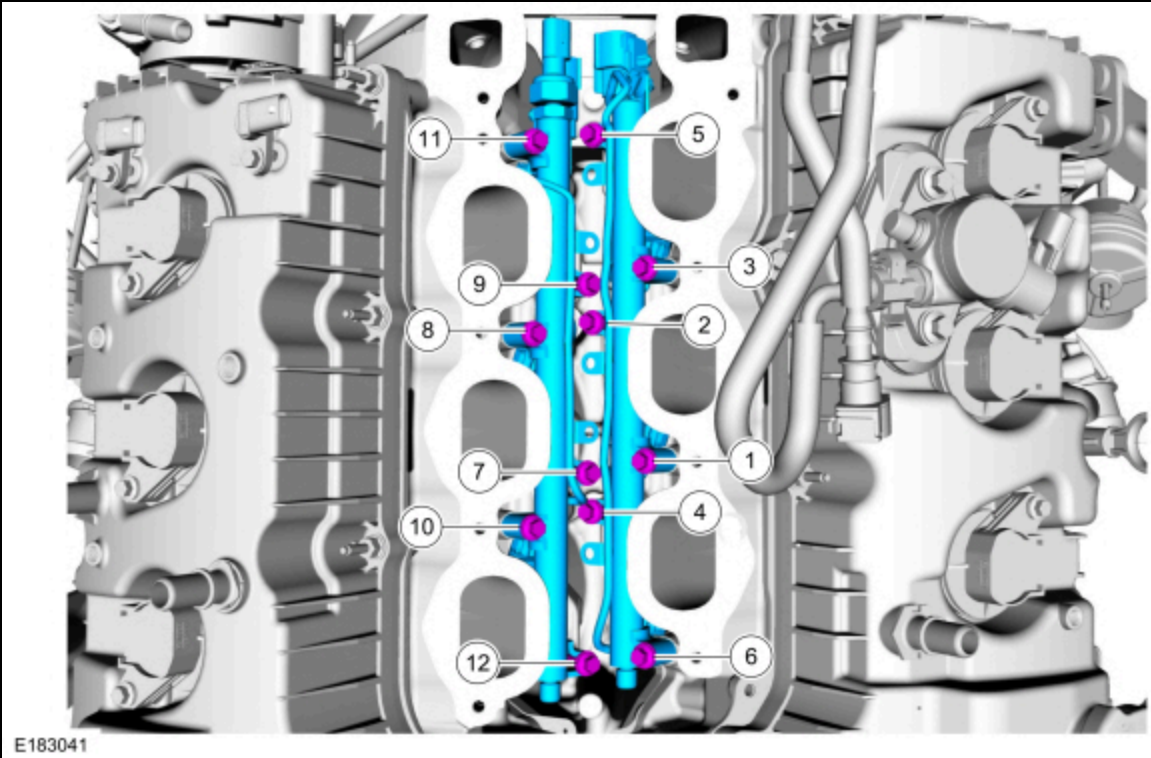
9. **NOTICE: Do not lubricate the new lower Teflon® fuel injector seals.**

NOTE: Make sure that new fuel rail bolts are installed.

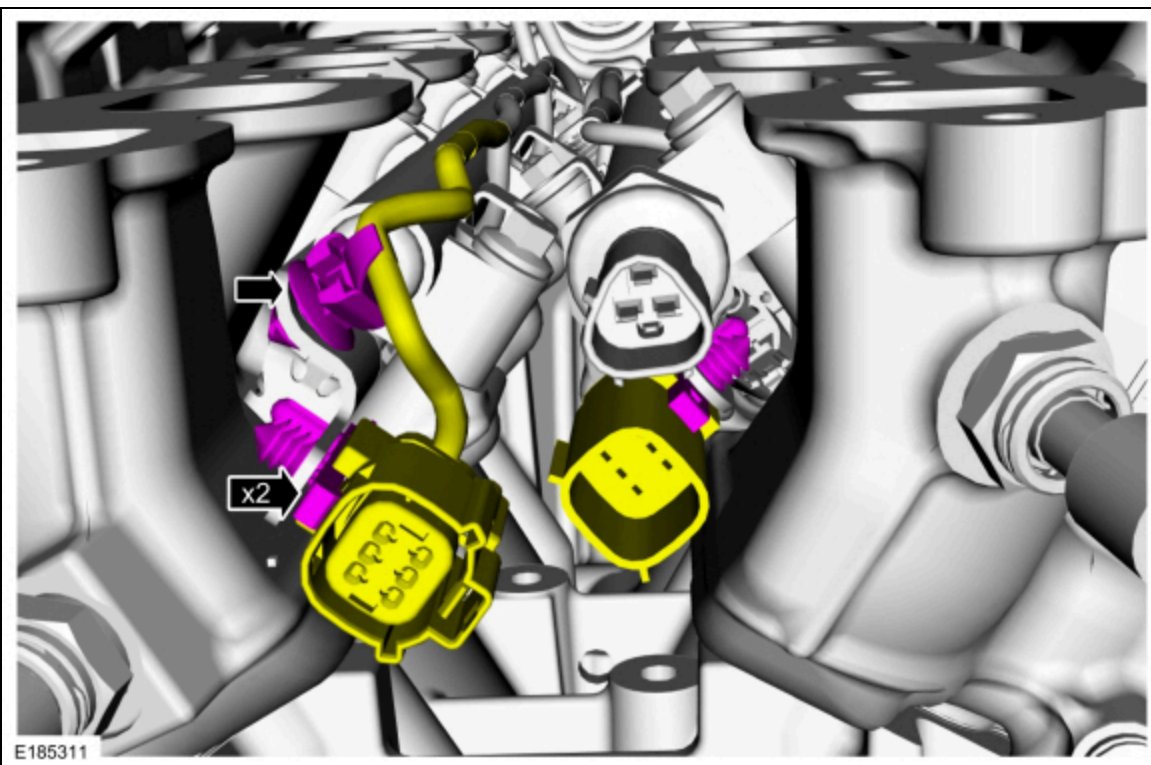
NOTE: Tighten the bolts in a method that draws the fuel rail evenly to the head, preventing a rocking motion.

Tighten the new fuel rail bolts in the following 3 stages.

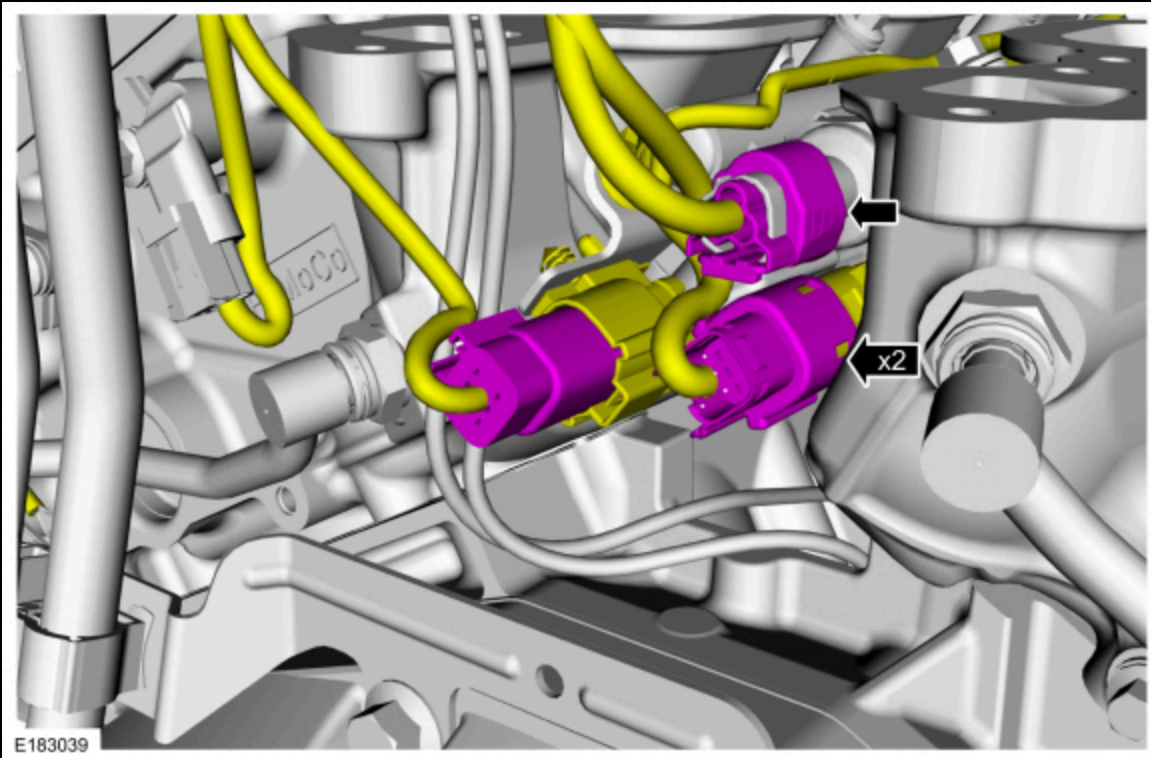
- Stage 1: Push down on the fuel rail face above the injectors.
- Stage 1: Tighten to 10 Nm (89 lb-in) in the sequence shown.
- Stage 2: Tighten an additional 45 degrees in the sequence shown.



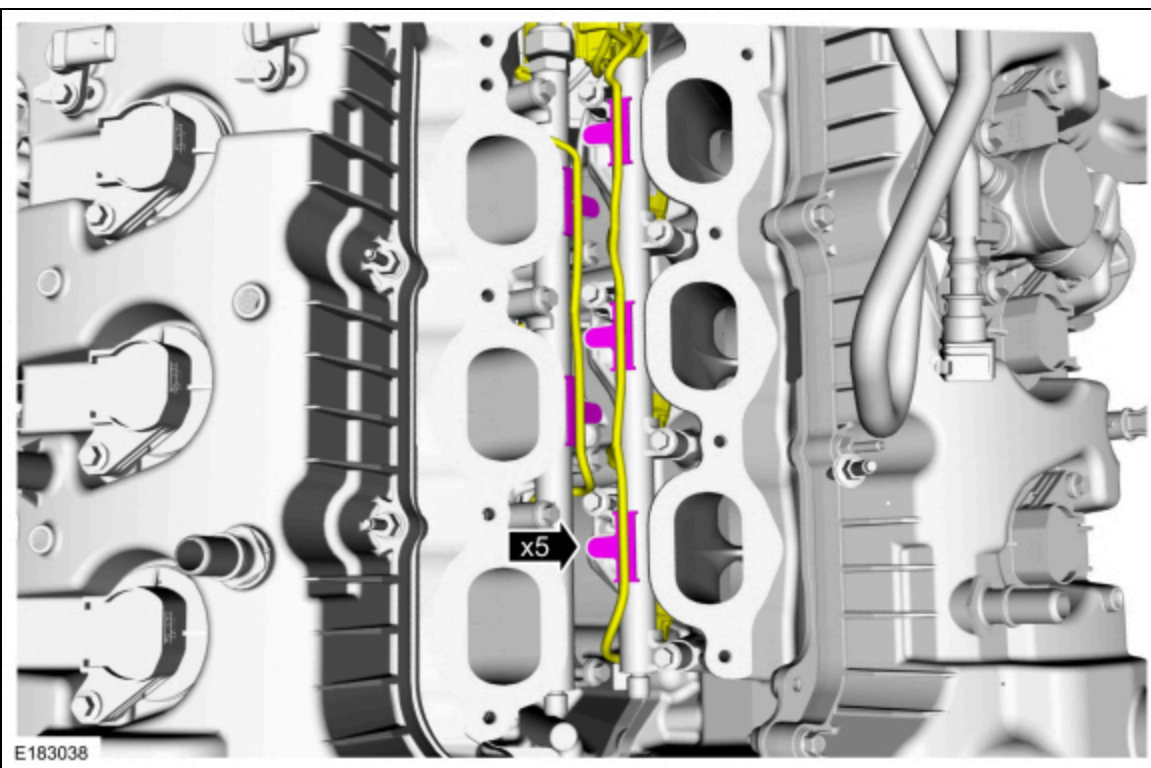
10. Attach the fuel injector wire harness electrical connector retainers and the fuel injector wire harness retainer to the fuel rail.



11. Connect the fuel injector wire harness connectors and the fuel rail pressure sensor wire connector.



12. Attach the fuel injector wire harness retainers to the fuel rail.



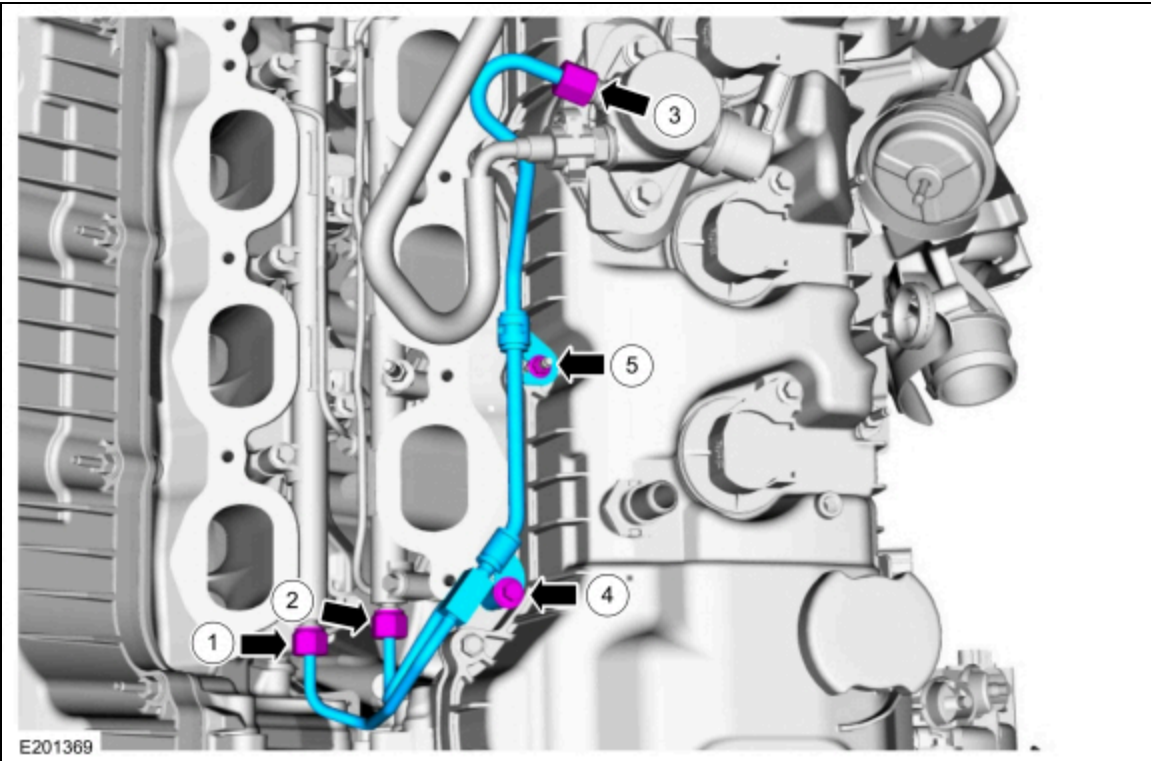
13. **NOTE:** Make sure that a new high pressure fuel tube, a new high pressure fuel tube bracket nut and a new high pressure fuel tube bracket bolt are installed.

NOTE: Install the new high pressure fuel tube, the high pressure fuel tube flare nuts, the new high pressure fuel tube bracket nut, the new bracket bolt and tighten finger tight at this stage. Tighten in the sequence shown.

NOTE: Calculate the correct torque wrench setting for the following torque. [Refer to the Torque Wrench Adapter Formulas in the Appendix.](#)

1. Tighten the high pressure fuel tube-to-RH fuel rail tube flare nut.
 - Stage 1: Tighten to 15 Nm (133 lb-in).
 - Stage 2: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.

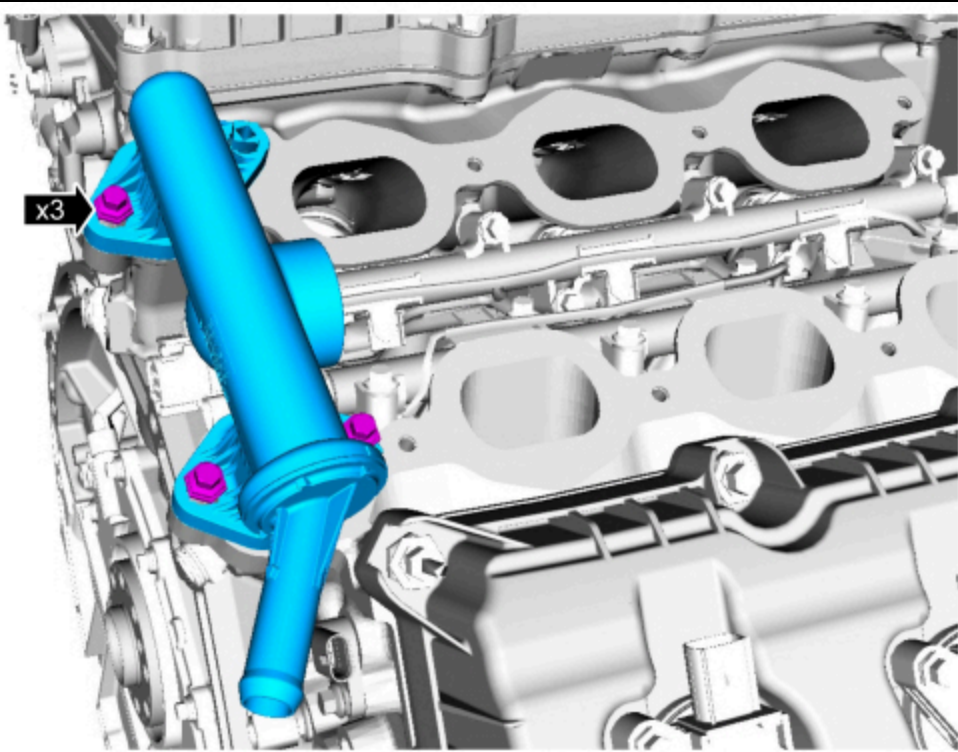
2. Tighten the high pressure fuel tube-to-LH fuel rail tube flare nut.
 - Stage 1: Tighten to 15 Nm (133 lb-in).
 - Stage 2: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.
3. Tighten the high pressure fuel tube-to-high-pressure fuel pump flare nut.
 - Stage 1: Tighten to 15 Nm (133 lb-in).
 - Stage 2: Tighten an additional 30 degrees or 50 Nm (37 lb-ft), whichever comes first.
4. Tighten the high pressure fuel tube bracket bolt.
 - Stage 1: Tighten to 10 Nm (89 lb-in).
 - Stage 2: Tighten an additional 45 degrees.
5. Tighten the high pressure fuel tube bracket nut.
 - Tighten to 8 Nm (71 lb-in).



14. **NOTE:** Use care when installing the fuel injection pump noise insulator. Spreading the openings will aid in installing the fuel injection pump noise insulator and reduce the risk of damage.

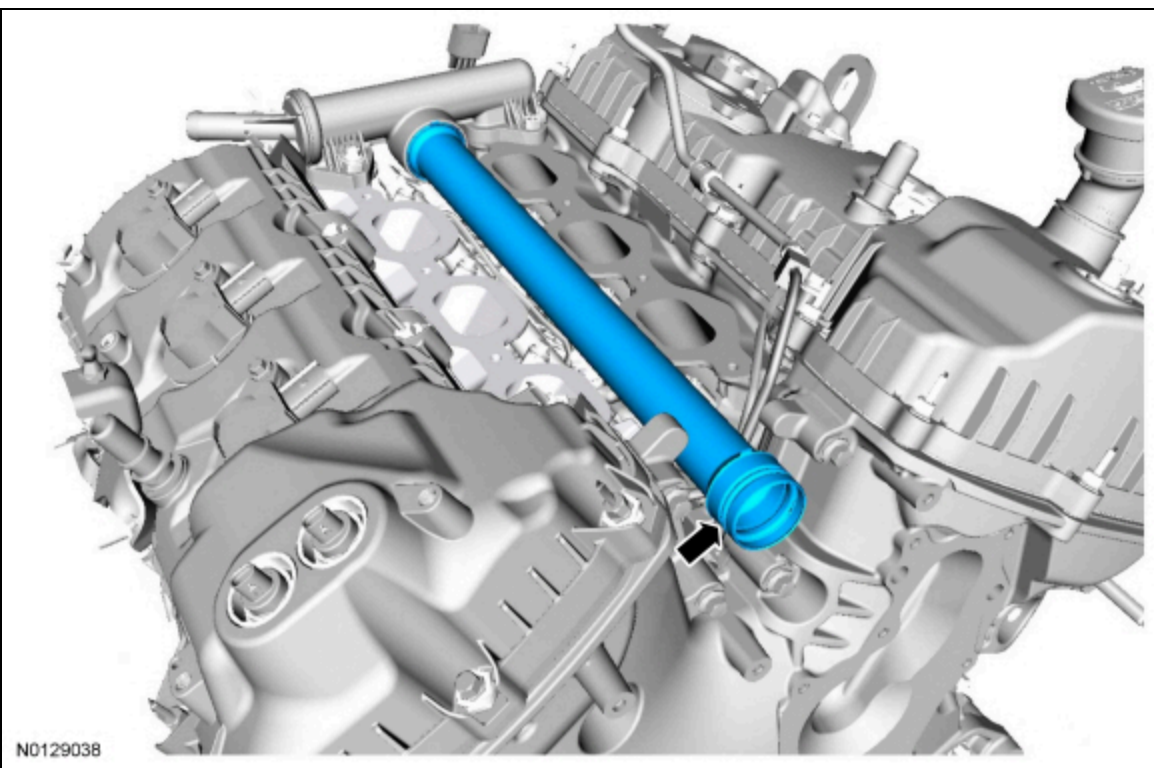
Install the fuel injection pump noise insulator shield.

15. Using new gaskets, position back the coolant crossover manifold assembly and install the 3 bolts.
 - Tighten to 10 Nm (89 lb-in) plus an additional 45 degrees.



16. **NOTE:** *Lubricate the O-ring seals with clean engine coolant.*

Using new O-ring seals, install the coolant inlet pipe.



17. Install the thermostat housing. For additional information, refer to Section 303-03.

18. Install the intake manifold. For additional information, refer to Section 303-01A.